

Benchmarking Hybrid Dense-Sparse Retrieval with Reranking for Academic Literature

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Abstract

Dense vector embeddings excel at understanding semantic themes, while sparse lexical models (BM25) excel at exact keyword matching. In this paper, we conduct a comprehensive benchmark comparing pure dense retrieval, sparse BM25 retrieval, and Reciprocal Rank Fusion (RRF) hybrid search combined with Cross-Encoder reranking across 20 academic domains. Our findings show that hybrid retrieval improves Recall@20 by 18.7% over pure dense search.

1. Motivation

Academic literature contains domain-specific nomenclature, math notation, and chemical names that out-of-the-box dense embedding models map poorly. Integrating sparse term frequency signals alongside dense cosine distance remedies this blind spot.

2. Proposed Hybrid Architecture

Our hybrid retrieval pipeline executes parallel searches: FAISS index retrieves top-50 dense candidates using sentence-transformers/all-MiniLM-L6-v2, while BM25 retrieves top-50 lexical candidates. Scores are merged via Reciprocal Rank Fusion: $RRF_score(d) = \sum (1 / (k + rank_i(d)))$. A cross-encoder reranker then scores the top-20 merged candidates to yield the final top-k context.

3. Results and Computational Overhead

Hybrid search achieved 96.2% Recall@10, significantly outperforming standalone BM25 (76.8%) and standalone Dense FAISS (88.4%). However, cross-encoder reranking added 45ms of inference latency per query, representing an engineering trade-off between speed and top-1 precision.

4. Advantages and Key Findings

1. Robustness: High retrieval quality across both keyword-heavy and conceptually ambiguous queries.
2. Reranking Accuracy: Cross-encoders capture deep token-level cross-attention between queries and retrieved passages.
3. Explainability: Transparent fusion of lexical token overlap and vector similarity.

5. Limitations

Increased computational complexity and memory requirements from maintaining dual indices (FAISS vector index + inverted BM25 index).

6. Conclusion

Hybrid search with reranking provides state-of-the-art retrieval accuracy for enterprise and academic knowledge synthesis pipelines.